

Quine - McClusky minimization

- # Tabular Method
- # When no. of inputs is too large
- # Large no of variables - 6
- # Computer driven

Example 1

$$F(A,B,C,D) = \sum m(0,1,3,7,8,9,11,15)$$

Step 0:

Write down the binary representation of the minterms.

	<u>Binary Representation</u>				Number of 1s (for next step)
	A	B	C	D	
m_0	0	0	0	0	0
m_1	0	0	0	1	1
m_3	0	0	1	1	2
m_7	0	1	1	1	3
m_8	1	0	0	0	1
m_9	1	0	0	1	2
m_{11}	1	0	1	1	3
m_{15}	1	1	1	1	4

Step 1:

Group the minterms based on number of 1's present in its binary representation.

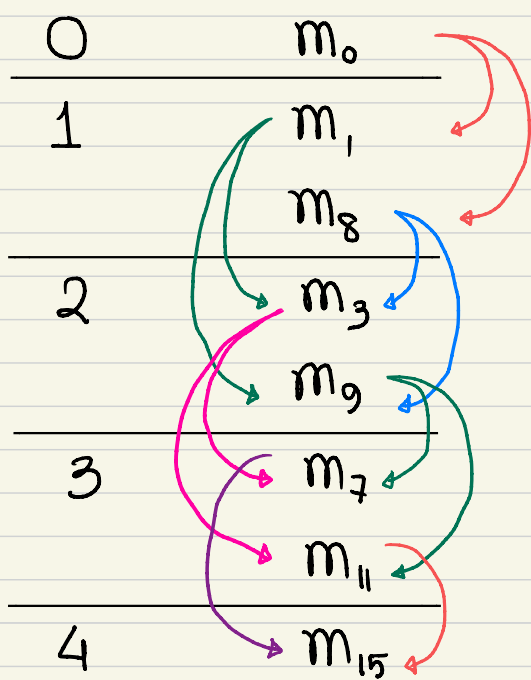
Group Minterm		Binary Representation				
		A	B	C	D	
0	m_0	0	0	0	0	✓
1	m_1	0	0	0	1	✓
	m_8	1	0	0	0	✓
2	m_3	0	0	1	1	✓
	m_9	1	0	0	1	✓
3	m_7	0	1	1	1	✓
	m_{11}	1	0	1	1	✓
4	m_{15}	1	1	1	1	✓

* Put a tick next to the 'group elements' which are part of valid pairs in the next step.

Step 2:

- * Check all possible pairs of consecutive group elements
- * Only consider the pairs of group elements which differ by exactly 1 bit.

Group Minterm



Check all pairs

Matching

$$\left. \begin{array}{cc|c|c} m_1 & 0 & 0 & 1 \\ m_3 & 0 & 1 & 1 \end{array} \right\} 00\underline{1}$$

- * differs by a single bit $\rightarrow$ valid
- * Write down matched bits 00 1.
- * Replace the dissimilar bit by '-'.
00 _ 1.
- * Put a tick next to m_1 and m_3 in previous table.
- * Try this for all other pair combination.

m_8 1 0 0 0
 m_3 0 0 1 1

X Not valid match

- * 3 bits are different
→ can't be considered.

Group	Matched pair	A B C D	
0	m_0, m_1	0 0 0 -	✓
	m_0, m_8	- 0 0 0	✓
1	m_1, m_3	0 0 - 1	✓
	m_1, m_9	- 0 0 1	✓
	m_8, m_9	1 0 0 -	✓
2	m_3, m_7	0 - 1 1	✓
	m_3, m_{11}	- 0 1 1	✓
	m_9, m_{11}	1 0 - 1	✓
3	m_7, m_{15}	- 1 1 1	✓
	m_{11}, m_{15}	1 - 1 1	✓

Group	Matched pair	A B C D		
0	m_0, m_1, m_8, m_9	- 0 0 -	}	$\bar{B}\bar{C}$
	m_0, m_8, m_1, m_9	- 0 0 -		
1	m_1, m_3, m_9, m_{11}	- 0 - 1	}	$\bar{B}D$
	m_1, m_9, m_3, m_{11}	- 0 - 1		
	m_3, m_7, m_{11}, m_{15}	- - 1 1	}	CD
	m_3, m_{11}, m_7, m_{15}	- - 1 1		

Prime Implicants

Repeat this step until no further valid matches found.

The unmatched elements (in the last or any other step) is considered to be Prime Implicants.

So, $\bar{B}\bar{C}$, $\bar{B}D$ and CD are prime implicants.

Now we have to determine the essential prime implicants.

Step 3 :

1. Make a table of prime implicants.
2. Prime Implicants along rows, minterms along columns.
3. For each P.I put a 'x' under each minterm that is covered (satisfied) by that PI.
4. Now visit each column, if the column contains a single 'x' then mark or circle it.
5. Finally include the minterms that contain at least one marked or circled 'x' in your solution.
(sum of products form)

Step 4

PI		0	1	3	7	8	9	11	15
$\bar{B} \bar{C}$	m_0, m_1, m_8, m_9	$\odot$	X			$\odot$	X		
$\bar{B} D$	m_1, m_3, m_9, m_{11}		X	X			X	X	
CD	m_3, m_7, m_{11}, m_{15}			X	$\odot$			X	$\odot$

✓

✓

$$F = \bar{B} \bar{C} + CD$$

Example 2

$$F(A,B,C,D) = \sum m(0,1,2,8,10,11,14,15)$$

Step 0:

	<u>Binary Representation</u>				Number of 1s (for next step)
	A	B	C	D	
m_0	0	0	0	0	0
m_1	0	0	0	1	1
m_2	0	0	1	0	1
m_8	1	0	0	0	1
m_{10}	1	0	1	0	2
m_{11}	1	0	1	1	3
m_{14}	1	1	1	0	3
m_{15}	1	1	1	1	4

Step 1:

Group Minterm		Binary Representation				
		A	B	C	D	
0	m_0	0	0	0	0	✓
1	m_1	0	0	0	1	✓
	m_2	0	0	1	0	✓
	m_8	1	0	0	0	✓
2	m_{10}	1	0	1	0	✓
3	m_{11}	1	0	1	1	✓
	m_{14}	1	1	1	0	✓
4	m_{15}	1	1	1	1	✓

Group	Minterm	Binary Representation				
		A	B	C	D	
0	$m_0 - m_1$	0	0	0	—	· X
	$m_0 - m_2$	0	0	—	0	✓
	$m_0 - m_8$	—	0	0	0	✓
1	$m_2 - m_{10}$	—	0	1	0	✓
	$m_8 - m_{10}$	1	0	—	0	✓
2	$m_{10} - m_{11}$	1	0	1	—	✓
	$m_{10} - m_{14}$	1	—	1	0	✓
3	$m_{11} - m_{15}$	1	—	1	1	✓
	$m_{14} - m_{15}$	1	1	1	—	✓

Group	Minterm	Binary Representation				
		A	B	C	D	
0	$m_0 - m_2 - m_8 - m_{10}$	—	0	—	0	} $\bar{B}\bar{D}$
	$m_0 - m_8 - m_2 - m_{10}$	—	0	—	0	
1	$m_{10} - m_{11} - m_{14} - m_{15}$	1	—	1	—	} AC
	$m_{10} - m_{14} - m_{11} - m_{15}$	1	—	1	—	

$$PI : \bar{B}\bar{D} + AC + \bar{A}\bar{B}\bar{C}$$

Step 3 :

Prime Implicants	0	1	2	8	10	11	14	15	
AC					x	(x)	(x)	(x)	✓
$\bar{B}\bar{D}$	x		(x)	(x)	x				✓
$\bar{A}\bar{B}\bar{C}$	x	(x)							✓

$$F = AC + \bar{B}\bar{D} + \bar{A}\bar{B}\bar{C}$$

Example 3 (With don't cares)

$$F(A,B,C,D) = \sum m(0,3,5,6,7,10,12,13) + \sum d(2,9,15)$$

Step 0: Include all terms (even don't cares).

	<u>Binary Representation</u>				Number of 1s (for next step)
	A	B	C	D	
m_0	0	0	0	0	0
m_2	0	0	1	0	1
m_3	0	0	1	1	2
m_5	0	1	0	1	2
m_6	0	1	1	0	2
m_7	0	1	1	1	3
m_9	1	0	0	1	2
m_{10}	1	0	1	0	2
m_{12}	1	1	0	0	2
m_{13}	1	1	0	1	3
m_{15}	1	1	1	1	4

Step 1:

Group Minterm		Binary Representation				
		A	B	C	D	
0	m_0	0	0	0	0	✓
1	m_2	0	0	1	0	✓
2	m_3	0	0	1	1	✓
	m_5	0	1	0	1	✓
	m_6	0	1	1	0	✓
	m_9	1	0	0	1	✓
	m_{10}	1	0	1	0	✓
	m_{12}	1	1	0	0	✓
3	m_7	0	1	1	1	✓
	m_{13}	1	1	0	1	✓
4	m_{15}	1	1	1	1	✓

Group Minterm		Binary Representation				
		A	B	C	D	
0	$m_0 - m_1$	0	0	—	0	✗ } $\bar{A}\bar{B}\bar{D}$
1	$m_2 - m_3$	0	0	1	—	✓
	$m_2 - m_6$	0	—	1	0	✓
	$m_2 - m_{10}$	—	0	1	0	✗ } $\bar{B}C\bar{D}$
1	$m_3 - m_7$	0	—	1	1	✓
	$m_5 - m_7$	0	1	—	1	✓
2	$m_5 - m_{13}$	—	1	0	1	✓
	$m_6 - m_7$	0	1	1	—	✓
	$m_9 - m_{13}$	1	—	0	1	✗ } $A\bar{C}D$
	$m_{12} - m_{13}$	1	1	0	—	✗ } $AB\bar{C}$
3	$m_7 - m_{15}$	—	1	1	1	✓
	$m_{13} - m_{15}$	1	1	—	1	✓

Group Minterm		Binary Representation				
		A	B	C	D	
0	$m_2 - m_3 - m_6 - m_7$	0	—	1	—	} $\bar{A}C$
	$m_2 - m_6 - m_3 - m_7$	0	—	1	—	
1	$m_5 - m_7 - m_{13} - m_{15}$	—	1	—	1	} BD
	$m_5 - m_{13} - m_7 - m_{15}$	—	1	—	1	

$$P.I = \bar{A}C + BD + \bar{A}\bar{B}\bar{D} + \bar{B}C\bar{D} + A\bar{C}D + AB\bar{C}$$

Step 3 :

exclude DONT CARE terms

Prime Implicants	Minterms	0	3	5	6	7	10	12	13	
$\bar{A}C$	2, 3, 6, 7		(X)		(X)	x				✓
BD	5, 7, 13, 15			(X)		x			x	✓
$\bar{A}\bar{B}\bar{D}$	0, 1	(X)								✓
$\bar{B}C\bar{D}$	2, 10						(X)			✓
$A\bar{C}D$	9, 13								x	
$AB\bar{C}$	12, 13							(X)	x	✓

$$F = \bar{A}C + BD + \bar{A}\bar{B}\bar{D} + \bar{B}C\bar{D} + AB\bar{C}$$